



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to V.V.D. PETRO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

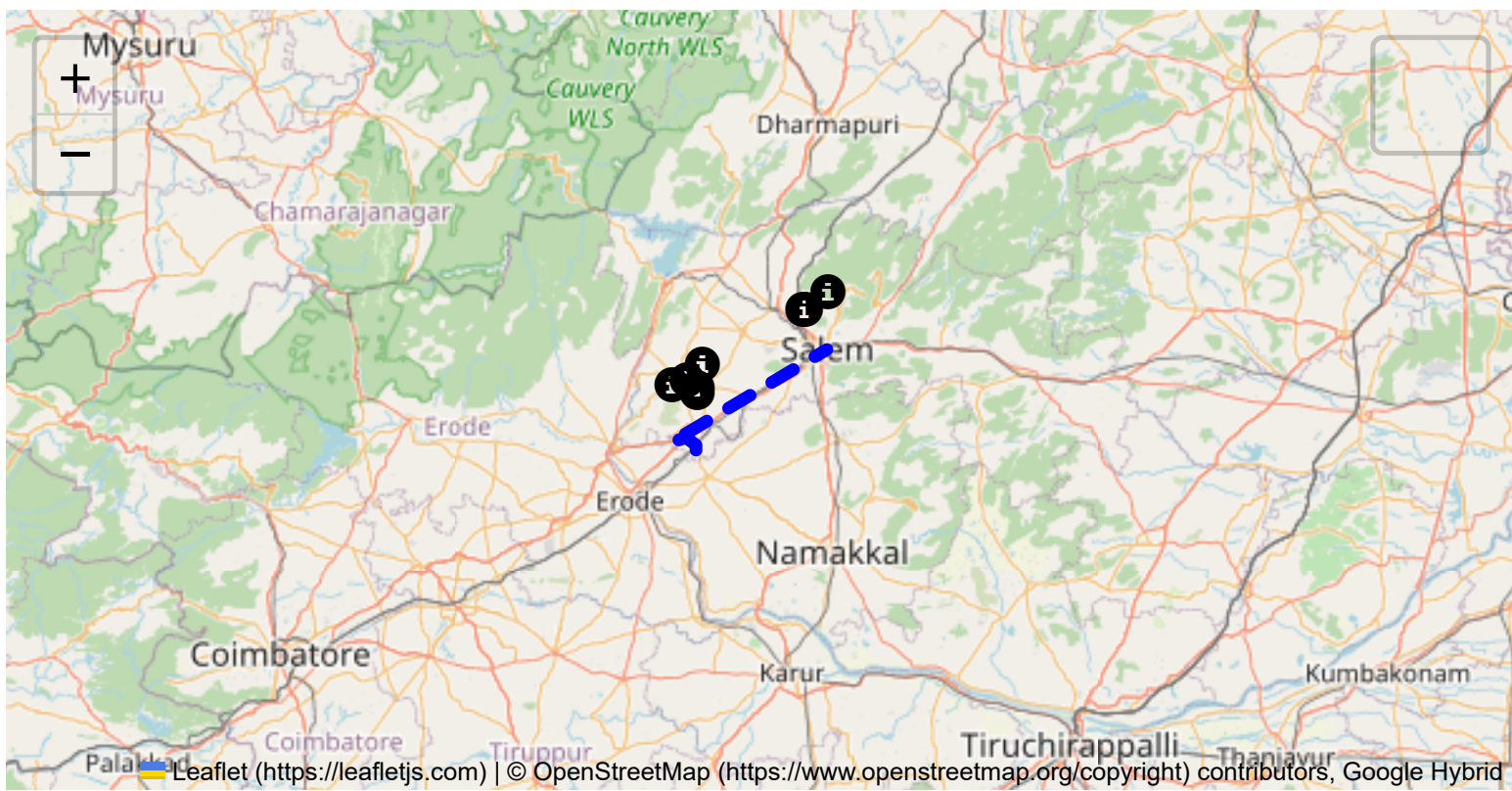
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

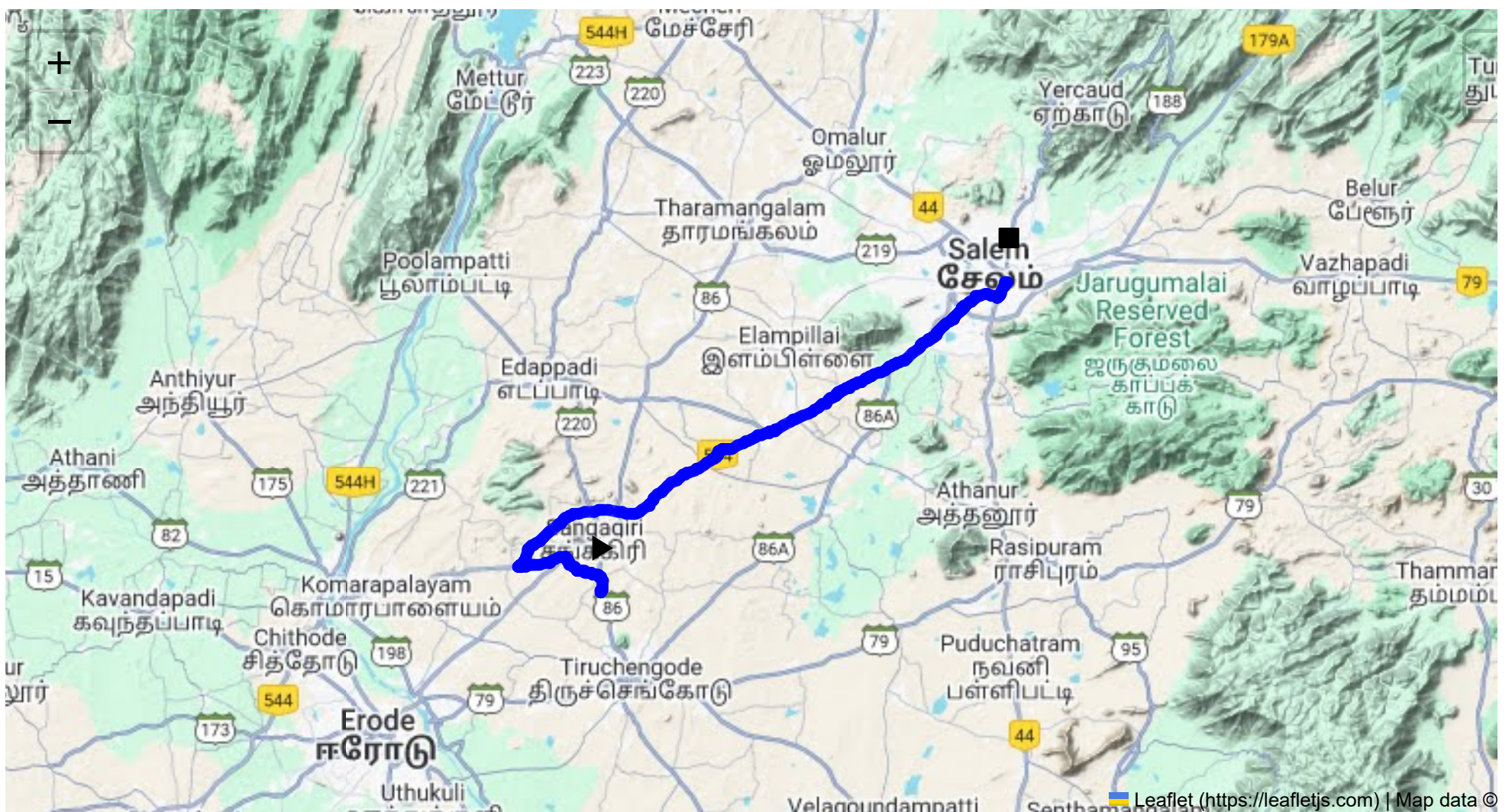
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 53.75 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.650399, 78.158669)



Welcome to the Journey Risk Management Study

Route Overview:

The route covers approximately 53.75 kilometers, starting from Sangagiri and ending in Salem, Tamil Nadu. It involves several critical waypoints, mainly major roads and highways, such as the Sankari RS - Erode Privu Road, Bhavani Main Road, and NH544. Being a primary route, it's likely to intersect with other major roads and could include both urban and semi-urban areas.

Typical Weather Conditions and Hazards:

- The region typically experiences a tropical climate, with hot summers and moderate rainfall during the monsoon season.
- Weather hazards include heavy rains causing flooding and reduced visibility, particularly between June and September.
- During summer (March to June), extremely high temperatures can lead to road surface damage and tire issues.

Traffic Patterns:

- Traffic congestion could be significant in urban sections, especially near intersections like NH544.
- Peak hours are usually morning (8 AM – 10 AM) and evening (5 PM – 8 PM) on weekdays.
- Markets and school zones along the route further increase congestion during specific time slots.

Road Quality and Infrastructure:

- NH544 is generally well-maintained, but smaller roads and certain stretches through urban areas may suffer from patchy maintenance.
- Potholes, uneven surfaces, and poorly marked intersections might pose challenges.

Alternative Routes:

- In case of emergencies, taking a detour via nearby highways or main roads like NH79 could be considered, although they might increase travel time.
- Local roads in the vicinity of these main highways may provide short bypass options but could have very limited capacity for heavy vehicles.

Local Regulations:

- Regulations on transporting hazardous materials include strict adherence to loading limits, proper vehicle markings, and obtaining necessary permits.
- Night-time restrictions might apply for transportation through densely populated areas.

Historical Incidents:

- Data may indicate sporadic incidents involving heavy vehicles, often due to speeding, brake failure, or poor road conditions, especially on hilly or sharp-curved sections.

Environmental Considerations:

- Sensitive areas include agricultural zones and residential sectors where care must be taken to prevent contamination.
- Waste management and spill response preparedness are critical in these regions.

Communication Coverage:

- Major towns and city areas typically have good communication coverage, but rural stretches and minor roads may have spotty service.
- Satellite communication tools could be advisable as a backup.

Emergency Response Times:

- Proximity to major towns like Erode and Salem shortens response times to an estimated 20-30 minutes in urban areas.
- Rural stretches or during adverse weather conditions could extend response times up to an hour.

Overall Risk Assessment:

- This route is moderately risky due to traffic congestion, variable road quality, and weather inconsistencies.
- Careful planning for timing, adherence to regulations, and preparation for adverse weather conditions can mitigate many of these risks.

Recommendations for Truck Drivers:

- Conduct regular vehicle checks, especially before embarking on routes with extreme temperature forecasts or post-rainy season.
- Stay updated on local traffic news and be prepared for detours during peak congestion times or in case of roadblocks.
- Have proper emergency kits that include communication devices, first aid, and tools for minor repairs.
- Additional training on recognizing and navigating through poorly maintained road sections and handling emergency situations might be beneficial.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
6	Turn	Medium	11.46048, 77.85036	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	High	11.61233, 78.10397	15 KM/Hr
12	Turn	Medium	11.61216, 78.10423	30 KM/Hr
13	Turn	Medium	11.64132, 78.15294	30 KM/Hr
14	Turn	High	11.64134, 78.15304	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	clinic	Salem Care Foundation	11.6371946, 78.1313713	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
19	hospital	Aravind Eye Hospital	11.641927, 78.1392347	30 km/h	Medium
20	police	Annadanapatti B 4 Police Station	11.6424031, 78.1404779	30 km/h	Medium
21	clinic	Shri Bharthi Nursing Home	11.6420069, 78.1424099	30 km/h	Medium
22	police	Police Control Room Police Station	11.6425701, 78.1459515	30 km/h	Medium
23	hospital	Sree Vasantham Hospital	11.6420424, 78.1463903	30 km/h	Medium
25	hospital	Anbu Hospital	11.6402823, 78.152387	30 km/h	Medium
26	clinic	Shanti Nursing Home, Salem	11.6407328, 78.1532635	30 km/h	Medium
27	clinic	Soundararajan Nursing Home	11.6379994, 78.1517239	30 km/h	Medium
28	clinic	Ramadas Nursing Home	11.6449413, 78.1545496	30 km/h	Medium
29	hospital	G.V.K. Hospital	11.6445919, 78.1555742	30 km/h	Medium
30	hospital	Life Care Hospital	11.6452364, 78.1546993	30 km/h	Medium
31	clinic	Sri Balaji Clinic	11.645189, 78.154708	30 km/h	Medium
32	hospital	Janani Sree Eye Hospital	11.6445844, 78.1543916	30 km/h	Medium
36	clinic	Ravi Eye Clinic	11.646684, 78.1547669	30 km/h	Medium
37	hospital	Vasan Eye Care Hospital, Gugai	11.6471454, 78.1559283	30 km/h	Medium
38	hospital	Pp Hospital	11.6474921, 78.1561723	30 km/h	Medium
39	hospital	Corporation Maternity Hospital	11.6524437, 78.154675	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
24	school	The gugai hr.sec school	11.6411104, 78.1472518	30 km/h	Medium
33	school	Veeralashmi Vidhyalaya Higher Secondary School	11.6414737, 78.1565634	30 km/h	Medium
34	school	Jayarani matric school	11.6463187, 78.1527477	30 km/h	Medium
35	school	Veeralakshmi Vidhyalaya Higher Secondary School	11.6422707, 78.1567216	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61233, 78.10397



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61216, 78.10423



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64132, 78.15294



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64134, 78.15304

